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import torch
from torch import nn

class FractionalConv(nn.Module):
    def __init__(self, alpha=0.8): # дробный
        порядок
        super().__init__()
        self.alpha = alpha
        self.conv = nn.Conv2d(64, 64, 3,
padding=1)

    def fractional_memory(self, x, history):
        # Интеграл по истории:  $\gamma(t-\tau)x(\tau)$ 
        memory_term = torch.mean(history,
dim=1, keepdim=True)
        return self.alpha * self.conv(x) + (1-
alpha) * memory_term

    def forward(self, x, history_buffer):
        out = self.fractional_memory(x,
history_buffer)
        return out, torch.cat([history_buffer[-10:],
```

out], dim=1)